HENRY JIANG

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Education

Georgia Institute of Technology *Atlanta, GA*

- B.S. in Computer Science | Concentrations: Artificial Intelligence, Computational Media *May 2026*
- GPA: 3.66 | Graduating with Highest Honors, Dean's List 2023, 2024, 2025, Eta Kappa Nu Scholar, IEEE Member

Skills

Languages: Python, Java, JavaScript, Typescript, C, C#, C++, HTML, CSS, R, SQL, Bash
Tools: Git, JUnit, Jira, Docker, Splunk, Firebase, Tensorflow, Pytorch, Scikit-learn, SciPy, MatplotLib, Seaborn, Pytorch Geometric, Pandas, Numpy, Splunk, Selenium, OpenCV, ASE, MATLAB
Frameworks: React, Next.js, Three.js, Node,js, REST APIs, Angular, Android Studio, Figma, Firebase, PostgreSQL, MySQL, MongoDB

Experience

Blender Foundation *Apr 2024 – Present*
Extension Developer & Core Contributor

- Designed and implemented 3 Blender addons using Python and Blender API, enhancing rigged animation workflow efficiency by 20+% for 1,000+ users by automating IK/FK snapping and dynamic pose switching
- Collaborated with senior developers via Gitea to resolve high-priority animation module bugs and optimize weight painting scripts, improving menu-latency by 15%

GT Fung Research Group *Aug 2023 – May 2026*
Undergraduate Research Assistant *Atlanta, GA*

- Developed 2 graph neural network (GNN) packages for crystal structure prediction achieving 90%+ accuracy on materials datasets with 10,000+ atomic configurations using PyTorch Geometric, ASE, and Pyswarms
- Solved inverse design bottleneck for unknown lattice parameters with a hybrid particle swarm optimization algorithm, achieving 80% structure match rate and reducing prediction error by 50% compared to baseline methods
- Engineered data preprocessing pipeline processing 500+ crystal structures from Materials Project database, discovering 10+ novel ground-state configurations through automated DFT validation

NASA *Jun 2024 – Aug 2024*
Information Systems Security Intern *Kennedy Space Center, FL*

- Engineered 4 production Splunk dashboards for Launch Control Center monitoring 100,000+ security events, implementing real-time alert systems with custom SPL queries and automated data aggregation pipelines
- Strengthened cybersecurity compliance by completing SSP and SCP revisions for 3 NE-XC sector teams with cross-collaboration, analyzing NIST/FISMA requirements and drafting 20+ PO&AMs for vulnerabilities
- Built automated data preprocessing pipeline for security event classification, achieving 75% reduction in false positive alerts through threshold optimization and time-series analysis techniques

Projects

DARWIN: Agentic Evolutionary GPT | OpenAI API, LLMs, Genetic Algorithms | *Github*

- Designed agentic evolutionary AI with self-improving agents optimizing NN architecture training code and hyperparameters achieving 1.2%+ MFU improvement and 2%+ perplexity improvement on fully trained models
- Engineered agentic CI/CD style debugging pipeline, persistent JSON memory system, and bidirectional communication interface for dynamic upgrade requests from agents (datasets, packages, scripts)

Mandelbrotian-Based Volatility Trading Model | *QuantConnect, Python, Scipy, Scikit-learn* | *Github*

- Trained probabilistic ML XGBoost models on 10+ technical features with scikit-learn, developing volatility-regime classifier with Hurst exponent and Hill estimator for tail-index tracking supplemented by RSI, Bollinger, ATR, and %B
- Backtested 50,000+ options contracts using QuantConnect, and validated with forward walk Monte Carlo simulations, reducing deep OTM option pricing error by ~22% leveraging Lévy-stable distributions

Viper: Artist Social Media with Image Poisoning | Postgres, React, MongoDB, Next.js, Node.js, AWS, Figma | *(wip)*

- Developed a full-featured social ecosystem with React, Next.js, and Node.js including real-time messaging, notification, and profile management, maintaining strict "No-LLM" development integrity and server-side image poisoning pipelines.
- Deployed a secure, cloud-native backend on AWS with PostgreSQL and MongoDB, managing complex state transitions and persistent user sessions across 10+ core feature modules.